Fiscal Year:	FY 2010	**Task Last Updated:**	FY 05/30/2012
PI Name:	Pawelczyk, Jim Ph.D.		
Project Title:	Improving Orthostatic Tolerance in Women: Control of Splanchnic and Cutaneous Vascular Capacitance		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	University, Park	**State:**	PA
Zip Code:	16801-6900	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	05/15/2004	**End Date:**	09/30/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date is now 9/30/2011, per C. Guidry/JSC (from 9/30/2010)--(ed. 10/26/2010) NOTE: End date is 9/30/2010, per C. Guidry/JSC (9/2010) NOTE: Received NCE to 9/30/2009 per J. Dardano/JSC (10/08) NOTE: Received NCE to 9/30/2008 per P. Denkins/JSC (10/07) NOTE: Received NCE to 9/30/2007 per PI (11/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Kenney, Larry (Pennsylvania State University) Leuenberger, Urs (Pennsylvania State College of Medicine) Williams, Nancy (Pennsylvania State University)		
Grant/Contract No.:	NNJ04HF45G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Clinical and empirical research reveals that orthostatic tolerance (OT) is less in women than men. This project seeks to relate sex differences in OT to biophysical properties of the circulation, and to validate physiological and pharmacological countermeasures to improve OT in women. In a gravitational field, cardiac filling is determined by the pressure difference between the right atrium and the hydrostatic indifferent point (HIP), the point in the circulation where pressure is unaffected by gravity. The HIP is the hydraulic equivalent of the center of mass of a solid object. We hypothesize that differences in anatomy and the distribution of vascular volume in women generate a HIP more caudal than men, causing greater falls in cardiac filling when standing. Accordingly, we will determine OT and the HIP in groups of women and men. Since the location of the HIP is dependent on regional vascular compliance, we will experimentally raise the HIP using a somatostatin analog or mechanical compression to reduce blood flow and vascular distension in the splanchnic circulation, while changing cutaneous compliance with passive skin cooling. These manipulations will produce measurable shifts in HIP, cardiac filling pressure, and OT. Moreover, they will establish the mechanism and means to ameliorate the greater incidence of post-spaceflight orthostatic intolerance and cardiovascular deconditioning in women by reducing vascular compliance in the two most distensible circulations in humans (splanchnic and cutaneous), shifting the HIP and vascular volume toward the heart. These new countermeasures can be adapted to spaceflight, and should prove applicable to populations of patients as well as orthostatically intolerant, but otherwise healthy, women.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	[Editor's note 5/30/2012: No Task Book report received. Progress section and Bibliography compiled from PI's NCE request in 2010.] Our final work is focused on completing data analysis and publication of the work resulting from this grant. Five publications funded all, or in part, from this work have been completed--see Bibliography. Additional papers are in various stages of review. Our work has shown that the segmental impedance response to orthostatic changes is non-linear, which has necessitated further refinement of our method to quantify the location of the volume indifferent point. New data analyses, using dynamic models of cardiovascular regulation developed during the past year, allow us to refine an approach to assess the extent of baroreflex engagement during orthostatic stress. Techniques can be utilized on all data collected to date and will allow us to assess the efficacy of countermeasures by quantifying the reduction in baroreflex engagement as orthostatic tolerance improves.
Bibliography Type:	Description: (Last Updated: 05/19/2015)
Articles in Peer-reviewed Journals	Jarvis SS, Pawelczyk JA. "Identification of the human electrical impedance indifferent point: a surrogate for the volume indifferent point?" Eur J Appl Physiol. 2009 Nov;107(4):473-80. Epub 2009 Aug 12. PubMed PMID: 19672616 , Nov-2009
Articles in Peer-reviewed Journals	Jarvis SS, Florian JP, Curren MJ, Pawelczyk JA. "Sex differences in vasoconstrictor reserve during 70 deg head-up tilt." Exp Physiol. 2010 Jan;95(1):184-93. Epub 2009 Sep 4. PubMed PMID: 19734174 , Jan-2010
Articles in Peer-reviewed Journals	Jarvis SS, Pawelczyk JA. "The location of the human volume indifferent point predicts orthostatic tolerance." Eur J Appl Physiol. 2010 May;109(2):331-41. Epub 2010 Jan 26. PubMed PMID: 20101407 , May-2010
Articles in Peer-reviewed Journals	Florian JP, Pawelczyk JA. "Non-esterified fatty acids increase arterial pressure via central sympathetic activation in humans." Clin Sci (Lond). 2009 Oct 2;118(1):61-9. PubMed PMID: 19426144 , Oct-2009
Articles in Peer-reviewed Journals	Florian JP, Pawelczyk JA. "Sympathetic and haemodynamic responses to lipids in healthy human ageing." Exp Physiol. 2010 Apr;95(4):486-97. Epub 2010 Jan 8. PubMed PMID: 20061355 , Apr-2010